

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
	(e)	(i)	Use of $r = \frac{v^2}{a}$ (1) Acceleration conversion i.e. /365/24/3600 (1) Answer = 8.8×10^{14} [m] or 8.8×10^{11} k[m] (1)			3	3	3	
		(ii)	Approximation used i.e. $D = \frac{r}{\theta}$ (1) Answer = 7.8×10^{14} [m] (1) Hence consistent (since overlap) i.e. valid conclusion based on calculations (1) Comparing e.g. 1.53 ± 0.15 and 1.73 ± 0.17 but also accept combined error = 20% or 8.8×10^{14} is less than 20% bigger than 7.8×10^{14} (1)			4	4	2	
			Question 7 total	4	6	10	20	9	0